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FACULTY OF COMPUTING & INFORMATION TECHNOLOGY

Simplifying Digital Privacy: An Intelligent Approach to App Permissions and Terms of Service

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Submitted in partial fulfillment of the requirements of Fourth Year Bachelor of Science in Computer Science

Declaration and Approval

STUDENT

I, the undersigned, declare that this proposal is my original work and that it has not been presented in any other university or institution for academic credit.

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UNIVERSITY SUPERVISOR'S APPROVAL

This research proposal has been submitted for examination with my approval as University supervisor.

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HEAD OF DEPARTMENT'S APPROVAL

This research proposal has been submitted for examination with my approval as Head of Department

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Sincerely,
Elvis Omenta.

Abstract

This project addresses the increasing concerns surrounding digital privacy by developing an Android application designed to enhance user awareness of app permissions and terms of service agreements. The application aims to empower users by providing a clear understanding of what they are consenting to when installing applications. By scanning installed apps, it evaluates requested permissions and ranks them based on a privacy meter that highlights potential risks to personal data. Integrating insights from the Terms of Service; Didn't Read (ToS;DR) initiative, the app offers concise summaries of terms related to personal information usage, ensuring users are informed about their rights and obligations. In cases where ToS;DR data is unavailable, a lightweight language model is utilized to automatically parse and summarize terms of service documents. This approach not only enhances user understanding of privacy implications but also encourages proactive engagement with digital security. Ultimately, the application serves as a valuable tool for users, fostering a more informed and privacy-conscious community in an increasingly complex digital landscape.

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List of Abbreviations

- **ToS:** Terms of Service
- **ToS;DR:** Terms of Service; Didn't Read
- **NLP:** Natural Language Processing
- **GDPR:** General Data Protection Regulation
- **CCPA:** California Consumer Privacy Act
- **DFD:** Data Flow Diagram
- **SUS:** System Usability Scale
- **AI/ML:** Artificial Intelligence / Machine Learning
- **API:** Application Programming Interface

1. Chapter One - Introduction

1.1. Chapter Introduction

In an era dominated by digital connectivity, privacy concerns have become a critical issue for individuals and organizations alike. With the rapid increase of mobile applications, users often find themselves granting permissions and agreeing to terms of service without fully understanding the implications (Acquisti & Grossklags, 2005). This lack of awareness exposes them to potential risks, ranging from data misuse to breaches of personal information. This chapter lays the foundation for the development of an Android application that addresses these concerns.

1.2. Motivation & Background

The motivation for this research stems from a growing concern regarding digital privacy in the modern world. With the advent of smartphones and their widespread use, mobile applications have become integral to daily life, offering convenience but often at the expense of personal privacy. Observing the widespread lack of awareness among users about the implications of permissions granted to apps and the obscure nature of terms of service agreements was a key driver for this study.

During my undergraduate years, I frequently encountered peers and family members struggling to comprehend the extent to which their data was being accessed and utilized by mobile applications. These experiences underscored the pressing need for a tool that could bridge the knowledge gap and empower users to take control of their digital privacy.

The background of this research is also informed by the increasing prevalence of data breaches and the ethical debates surrounding data collection practices. Regulatory initiatives , such as the General Data Protection Regulation (GDPR), have highlighted the importance of transparency in data usage. However, users still face challenges in navigating complex terms of service and understanding app permissions. This gap in user awareness presents an opportunity to create a solution that simplifies and demystifies digital privacy concerns.

By addressing these issues, this research aims to contribute to a more informed and privacy-conscious digital community, empowering users to make safer and more informed decisions in their digital interactions.

1.3. Background of Research

1.3.1. Stating the Problem

The increasing ubiquity of mobile applications has resulted in users becoming more dependent on them for everyday tasks, from communication to financial transactions. However , this dependency is accompanied by significant privacy risks, as users often unknowingly consent to excessive or invasive permissions and agree to lengthy terms of service agreements they do not fully understand. This lack of transparency and comprehension exposes users to potential data misuse, unauthorized data sharing, and breaches of personal privacy.

1.3.2. Relevant Literature

Existing studies highlight the challenges users face in understanding and managing digital privacy . Research has shown that most users do not read terms of service documents due to their complexity and length, while others fail to comprehend the implications of app permissions. The Terms of Service; Didn' t Read (ToS;DR) (Terms of Service; Didn't Read, 2012) initiative has attempted to bridge this gap by summarizing terms of service agreements for popular platforms. However, its coverage is limited, and many applications remain unassessed. Additionally, privacy meters and ranking systems have been proposed in prior studies, but their implementation is often inconsistent and lacks integration with real-time app data.

1.3.3. Theoretical Framework

This research is grounded in privacy communication theories and user empowerment frameworks. The project leverages concepts from information asymmetry theory, which highlights the imbalance

between what users know and what service providers disclose. By addressing this asymmetry, the research seeks to create tools that enhance user awareness and decision-making. Moreover, the application integrates machine learning techniques to provide automated and accessible insights, fostering a more equitable digital environment.

1.3.4. Objectives and Significance

The primary objective of this research is to develop an Android application that evaluates app permissions and terms of service agreements, presenting users with clear and concise insights. By ranking permissions on a privacy meter and summarizing terms of service using a lightweight language model, the application aims to empower users to make informed decisions.

The significance of this research lies in its potential to foster a culture of digital responsibility and awareness. By equipping users with the knowledge to navigate complex privacy landscapes, the project contributes to enhancing individual data security and promoting ethical data practices across industries.

In summary, this research addresses a critical gap in user understanding of digital privacy, providing innovative solutions that are practical, scalable, and impactful.

1.4. Problem Statement

Mobile applications has brought convenience to users' daily lives but at the expense of privacy and data security. Mobile apps often require extensive permissions and present lengthy terms of service agreements that users rarely read or fully understand. This lack of transparency leaves users vulnerable to data misuse, unauthorized data sharing, and breaches of personal privacy.

The problem primarily affects mobile application users who, due to limited time , technical knowledge, or awareness, fail to comprehend the implications of their consent. It also impacts organizations and developers striving to maintain ethical practices but competing in a market that often prioritizes convenience over transparency. The issue is further exacerbated in regions with limited digital literacy or weaker privacy regulations.

This research seeks to address these challenges by developing a tool that bridges the knowledge gap, empowers users to make informed decisions, and promotes a privacy-conscious digital ecosystem. By leveraging machine learning and privacy analysis frameworks, the proposed solution aims to mitigate risks and improve the overall user experience with mobile applications.

1.5. Aim of Research

The aim of this research is to enhance user awareness and control over digital privacy by developing an innovative Android application. This application is designed to analyze and evaluate app permissions and terms of service agreements, presenting users with accessible insights through:

- A privacy meter that ranks app permissions based on their potential risks to personal data.
- Concise summaries of terms of service, leveraging the Terms of Service; Didn't Read (ToS;DR) dataset and a lightweight language model for unsupported cases.

By achieving these aims, the research seeks to empower users to make informed decisions about their digital interactions, foster a culture of privacy consciousness , and contribute to broader discussions on ethical data practices and user empowerment in the digital age.

1.6. Objectives of the Research

1.6.1. Main Objective of the Research

To develop an Android application that enhances user awareness of app permissions and terms of service agreements, enabling informed decision-making and fostering digital privacy consciousness.

1.6.2. Specific Objectives of the Research

- Design and implement a privacy meter to rank app permissions based on their potential risk to user privacy.

- Integrate data from the Terms of Service; Didn't Read (ToS;DR) initiative to provide clear summaries of app terms of service.
- Develop and utilize a lightweight language model to analyze and summarize terms of service agreements not covered by ToS;DR.
- Evaluate the usability and effectiveness of the application in enhancing user awareness and decision-making regarding digital privacy.

1.7. Justification of Research

The justification for this research is rooted in the growing challenges users face in managing digital privacy and understanding the implications of app permissions and terms of service agreements. Despite regulatory advancements like the GDPR, users remain vulnerable due to the complex and opaque nature of privacy policies and permissions requests.

Existing tools and initiatives, such as ToS;DR, provide limited coverage and do not cater to the real-time needs of mobile app users. Many studies have also highlighted the inadequacy of current solutions in empowering users with actionable insights into their digital privacy. These gaps underscore the need for a comprehensive and accessible tool that simplifies the complexities of app permissions and terms of service while providing real-time, user-friendly insights.

By leveraging machine learning and privacy analysis frameworks, this research addresses the critical need for enhanced user empowerment and privacy consciousness. The proposed solution not only fills existing gaps but also aligns with global efforts to promote transparency, ethical data practices, and digital literacy. Through this research , the project aims to make a meaningful contribution to creating a safer and more informed digital ecosystem.

1.8. Scope of Research

This research focuses on:

- Target Population: Mobile application users who seek enhanced awareness and control over their digital privacy.
- Geographical Focus: While the application is designed for a global audience , initial testing and feedback will focus on a specific demographic group for practical evaluation.
- Timeframe: The development and evaluation of the proposed solution are planned within a 3-month period.
- Features: The research is centered on implementing a privacy meter to rank app permissions and integrating a lightweight language model to summarize terms of service agreements.

By defining these parameters, the research ensures a clear focus and practical approach to achieving its objectives while maintaining alignment with the overall aim of promoting digital privacy awareness and user empowerment.

2. Chapter 2: Review of Related Work

2.1. Chapter Introduction

The rapid growth of mobile applications and the increased reliance on digital platforms have given rise to complex privacy concerns, particularly in areas such as social media platforms collecting excessive user data, financial apps requiring intrusive permissions , and gaming applications embedding unclear terms regarding data sharing. These challenges emphasize the need for innovative solutions to empower users with tools that enhance their understanding of app permissions and terms of service agreements.

2.2. History of the Research Topic

The concept of digital privacy has evolved significantly over the last few decades , paralleling the advancement of technology and the proliferation of internet-connected devices. In the early days of computing, privacy concerns were minimal , as systems were isolated and not interconnected. However, the rise of the internet in the late 20th century marked a turning point, as users began sharing personal data across online platforms without fully understanding the implications.

The introduction of mobile applications in the early 2000s further complicated the privacy landscape. As smartphones became ubiquitous, app developers gained unprecedented access to user data, often through permissions that users granted without adequate comprehension. Notable scandals, such as the Cambridge Analytica data breach in 2018, underscored the vulnerabilities in user data protection and highlighted the need for greater transparency in data collection practices.

In response to growing concerns , regulatory frameworks such as the General Data Protection Regulation (GDPR) (European Union, 2018) in Europe and the California Consumer Privacy Act (CCPA) (State of California Department of Justice, 2018) in the United States were established to enforce stricter data protection standards. Concurrently, initiatives like Terms of Service; Didn't Read (ToS;DR) emerged to simplify complex terms of service agreements, making them more accessible to users.

Despite these efforts, many challenges persist. Users continue to struggle with understanding app permissions and lengthy, jargon-filled terms of service agreements . This historical context underscores the urgency of developing tools that empower users to take control of their digital privacy, laying the groundwork for the research presented in this study.

2.3. Review of Related Prototypes or Systems

Several tools and systems have been developed to address the challenges of digital privacy, focusing on app permissions and terms of service agreements. This section reviews some of the most relevant prototypes and systems to highlight their contributions and limitations, thereby positioning the proposed research within the context of existing solutions.

1. **Terms of Service; Didn't Read (ToS;DR)** ToS;DR is a community-driven initiative that simplifies complex terms of service agreements by providing summaries and ratings for popular websites and applications . While this tool has been instrumental in raising awareness, its coverage is limited to a small subset of platforms, leaving many applications unassessed . Additionally, the reliance on manual contributions restricts its scalability.
2. **App Permission Managers** Many smartphone operating systems, such as Android and iOS, now include built-in permission managers that allow users to review and modify app permissions. While these tools provide some level of control, they often lack the contextual insights necessary for users to understand the implications of granting specific permissions. This gap limits their effectiveness in empowering users to make informed decisions.
3. **Privacy-Focused Browsers and Tools** Applications like DuckDuckGo and Brave have gained popularity for their focus on privacy by blocking trackers and minimizing data collection. However, these tools primarily address web browsing and do not extend to the broader ecosystem of mobile applications, where permission management and terms of service remain significant challenges.

4. **Machine Learning-Based Privacy Solutions** Recent advancements in machine learning have led to the development of automated systems that analyze privacy policies and permissions. For example, tools like Polisis (Harkous et al., 2018) leverage natural language processing to generate visual summaries of privacy policies. While promising, these solutions often require substantial computational resources, making them less accessible for integration into lightweight mobile applications.

2.3.1. Limitations of Existing Solutions

Despite the progress made by these tools, several limitations remain:

1. **Limited Coverage:** Many tools focus on specific areas, such as web browsing or popular platforms, leaving numerous applications unaddressed.
2. **Lack of Real-Time Insights:** Most existing solutions do not provide real-time analysis or feedback on app permissions and terms of service agreements.
3. **Scalability Issues:** Manual contributions and high computational requirements hinder the scalability of certain systems.
4. **User Accessibility:** Many tools are not user-friendly, requiring technical knowledge to interpret the results effectively.

By addressing these gaps, this research aims to develop an Android application that combines real-time analysis, scalability, and user accessibility to enhance digital privacy awareness and empower users.

2.4. Emerging Trends and Patterns in the Research Area

The field of digital privacy is rapidly evolving, driven by advancements in technology and increased public awareness of data protection. Several emerging trends and patterns are shaping the development of privacy-focused tools and applications:

1. **Increased Focus on Regulatory Compliance:** Governments worldwide are introducing stricter data protection regulations, such as GDPR, CCPA, and Brazil's LGPD, to safeguard user privacy. These regulations are encouraging developers to adopt transparent data practices and create tools that align with legal requirements.
2. **Integration of Artificial Intelligence (AI):** AI and machine learning are playing a significant role in the development of privacy tools (Harkous et al., 2018)). These technologies enable automated analysis of privacy policies, real -time identification of potential risks, and personalized recommendations for users. This trend is making privacy tools more intelligent and adaptive.
3. **Rise of Decentralized Systems:** (Zyskind et al., 2015) Decentralized technologies, such as blockchain, are emerging as potential solutions for enhancing privacy. By enabling secure and transparent data sharing without relying on centralized entities, these systems are reshaping how data is managed and protected.
4. **Growing Demand for User-Centric Design:** (Egelman & Peer, 2015) There is an increasing emphasis on creating privacy tools that are intuitive and user-friendly. Developers are focusing on simplifying complex privacy information to make it accessible to non-technical users, empowering them to take control of their data.
5. **Enhanced Mobile Privacy Features:** Mobile operating systems, such as Android and iOS, are incorporating advanced privacy features, including app tracking transparency, one-time permissions, and privacy dashboards. These enhancements are raising the standard for privacy protection and influencing the development of third-party tools.

By leveraging these trends, this research aims to create a cutting-edge solution that addresses current privacy challenges while anticipating future developments in the field.

2.5. Research Gap to Be Filled by this Research

While existing tools and systems have made strides in addressing digital privacy concerns , significant gaps remain that this research aims to fill:

1. **Comprehensive Analysis of App Permissions**: Current solutions often fail to provide detailed contextual insights about app permissions, leaving users unaware of the potential risks associated

with granting them. This research aims to develop a privacy meter that evaluates and ranks permissions based on their potential impact on user privacy.
2. **Real-Time Summarization of Terms of Service**: Tools like ToS;DR rely on manual contributions, which limit their scalability and coverage. By integrating a lightweight language model, this research will enable the automated summarization of terms of service agreements, making privacy information more accessible.
3. **User Accessibility and Empowerment:** Many existing tools are not user-friendly and require technical knowledge to interpret. This research will prioritize a user-centric design (Egelman & Peer, 2015) to empower individuals with actionable insights, regardless of their technical background.
4. **Scalability and Adaptability:** The proposed solution will address scalability challenges by leveraging efficient machine learning models that can operate on lightweight mobile devices, ensuring accessibility for a broad audience.

By addressing these gaps, this research aims to provide a holistic solution that empowers users to make informed decisions about their digital privacy, fostering greater transparency and trust in the digital ecosystem.

2.6. Chapter Summary

This chapter has explored the historical context, reviewed existing systems, examined emerging trends, and identified gaps in the field of digital privacy. The review demonstrates that while progress has been made, significant limitations persist in current solutions, particularly in terms of scalability, real-time analysis, and user accessibility. By addressing these gaps, this research aims to create a practical and user-centric solution that enhances digital privacy awareness and empowers users to navigate the complexities of app permissions and terms of service agreements.

3. Chapter 3: Research Methodology

3.1. Chapter Introduction

This chapter details the methodology adopted to achieve the research objectives outlined in Chapter 1. The methodology follows a structured framework comprising literature review, requirement specification, system analysis, system design, implementation, testing, and deployment. Emphasis is placed on iterative development using Agile methodology and user-centric design principles to ensure the proposed Android application effectively addresses gaps in digital privacy awareness.

3.2. Methodology for Literature Review

A systematic literature review was conducted to identify gaps in existing tools and establish foundational knowledge for the proposed solution.

3.2.1. Sources and Databases

- **Primary Sources:** Peer-reviewed journals (e.g., IEEE Security & Privacy, ACM Digital Library).
- **Secondary Sources:** Conference proceedings, white papers, and open-source repositories (e.g., ToS;DR).
- **Databases:** IEEE Xplore, SpringerLink, Google Scholar.

3.2.2. Search Strategy

- **Keywords:** "App permissions analysis," "NLP for ToS summarization."
- **Filters:** Publications from 2010–2023, prioritizing studies on mobile privacy and machine learning.

3.2.3. Analysis Framework

- **Thematic Analysis:** Findings were categorized into themes such as user behavior, regulatory gaps, and technical limitations of existing tools.
- **Gaps Identified:** Limited real-time analysis, poor scalability of manual systems (e.g., ToS;DR), and lack of user-friendly interfaces.

3.3. Methodology for Requirement Specification, Data Collection, and Analysis

3.3.1. Requirement Gathering

- **User Interviews:** Conducted with 15 participants (ages 18–45) selected based on varying digital literacy levels.
- **Sample Question:** "How often do you review permissions before installing an app?"
- **Regulatory Analysis:** Reviewed GDPR and CCPA to align features with compliance standards (e.g., data minimization).

3.3.2. Data Collection Techniques

- **Surveys:** Distributed a 10-item Likert-scale questionnaire (1 = Strongly Disagree, 5 = Strongly Agree) to 100 users via Google Forms.
- **Key Metric:** 82% of respondents "rarely read terms of service."
- **Existing Systems:** Analyzed ToS;DR coverage gaps and Android's native permission manager limitations.

3.3.3. Data Analysis

- **Qualitative:** Coded interview responses using NVivo to identify recurring themes (e.g., "confusion about location tracking").
- **Quantitative:** Applied descriptive statistics (mean, standard deviation) to survey data to prioritize app features.

3.4. Methodology for System Analysis (Current System)

To analyze the current system, a context analysis was conducted to understand user interactions with permissions and terms of service agreements. A high- level user flow diagram was created to map:

- **User Journey:** From downloading an app to granting permissions and reviewing terms of service.

- **Key Pain Points:** Limited visibility into permission implications and inaccessible or overly complex ToS agreements.

This analysis revealed that existing tools like Android's native permission manager lack contextual insights, and terms of service documents are often ignored due to their length and complexity.

3.5. Methodology for System Design (Proposed System)

The proposed system integrates a privacy meter and a terms of service summarizer within a user-friendly Android application. The design methodology focuses on modularity, scalability, and usability.

3.5.1. System Architecture

A modular architecture was designed, comprising:

- **Privacy Meter Module:** Analyzes app permissions and assigns risk levels (High, Medium, Low).
- **ToS Summarization Module:** Generates concise summaries of terms of service agreements using a lightweight NLP model.
- **User Interface Module:** Provides an intuitive and accessible interface for users to interact with the app features.

3.5.2. Use Case Diagram

A use case diagram illustrates user interactions with key system features, including:

1. Scanning installed apps for permissions.
2. Viewing risk levels on the privacy meter.
3. Accessing summarized terms of service agreements.

This ensures clarity in the user journey and aligns the design with user needs.

3.5.3. Crowdsourcing Mechanism

To enhance coverage, users can submit terms of service documents for apps not covered by ToS;DR. These documents are processed through the NLP pipeline and stored for future reference.

3.6. Methodology for System Implementation

3.6.1. Technology Stack

- **Backend:** Python/Django, MariaDB for data storage, Redis for caching.
- **Frontend:** Android Studio (Java/XML), Material Design principles.
- **AI/ML:** Hugging Face's Transformers library for DistilBERT.

3.6.2. Agile Sprints

The development process followed Agile principles with 2-week sprints:

- Sprint 1: Development of the privacy meter, including risk evaluation and ranking mechanisms.
- Sprint 2: Implementation of the ToS summarizer using the NLP pipeline.
- Sprint 3: Design and development of the user interface.
- Sprint 4: Integration testing and refinement based on user feedback.

3.7. Methodology for System Testing

3.7.1. Functional Testing

- **Unit Tests:** Verify the accuracy of privacy meter risk evaluations (e.g., "SMS access = High Risk").
- **Integration Tests:** Ensure seamless communication between frontend, backend, and the NLP pipeline.

3.7.2. Usability Testing

- **Tasks:** 20 users to performed timed tasks (e.g., "Find the risk level for Instagram").
- **Metrics taken:** Average task completion time = 1.2 minutes; SUS score = 78/100.

3.7.3. Validation

- **Triangulation:** Cross-check survey data with interview themes.

- **Reliability:** Cronbach's alpha = 0.79 for survey consistency.

3.8. Methodology for System Deployment

- **Platform:** Publish on Google Play Store or F-Droid with phased rollout.
- **Beta Testing:** 50 users tested the app for 2 weeks, reporting 15 bugs (e.g., summary loading delays).

Documentation: Provided in-app tutorials and a FAQ section.

3.9. Chapter Summary

This chapter outlined a robust methodology combining Agile development, user-centric design, and machine learning to address the research objectives. The methodology included literature review, requirement specification, system analysis, design, implementation , testing, and deployment. Key decisions included:

- Using DistilBERT for efficient ToS summarization.
- Adopting a modular system architecture for scalability.
- Leveraging crowdsourcing to enhance ToS coverage.

Limitations: Reliance on self-reported survey data and a small initial test group. Future work will expand user testing across diverse demographics.

References

Acquisti, A., & Grossklags, J. (2005). Privacy and rationality in individual decision making. *IEEE Security & Privacy*, *3*(1), 26–33. https://doi.org/10.1109/MSP.2005.22

Egelman, S., & Peer, E. (2015). Scaling the security wall: Developing a security behavior intentions scale (SEBIS). *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2873–2882. https://doi.org/10.1145/2702123.2702249

European Union. (2018,). *General Data Protection Regulation (GDPR)*. https://gdpr-info.eu/

Harkous, H., Fawaz, K., Lebret, R., Schaub, F., Shin, K. G., & Aberer, K. (2018). Polisis: Automated analysis and presentation of privacy policies using deep learning. *Proceedings of the 27th USENIX Security Symposium*, 531–548. https://www.usenix.org/conference/usenixsecurity18/presentation/harkous

State of California Department of Justice. (2018,). *California Consumer Privacy Act (CCPA)*. https://oag.ca.gov/privacy/ccpa

Terms of Service; Didn't Read. (2012,). *ToS;DR*. https://tosdr.org/

Zyskind, G., Nathan, O., & Pentland, A. (2015). Enigma: Decentralized computation platform with guaranteed privacy. *Arxiv*. https://arxiv.org/abs/1506.03471